healthcare

MDPI

Article

Cataract Surgery in Low-Income Countries: A Good Deal!

Steffen Flessa

Department of Health Care Management, Faculty of Law and Economics, University of Greifswald, 17487 Greifswald, Germany; steffen.flessa@uni-greifswald.de; Tel.: +49-3834-420-24-77

Abstract: Cataract is a major cause of blindness worldwide. In particular, in low-income countries, the burden of disease as well as its direct and indirect economic cost are a major challenge for the population and economy. In many cases, it would be possible to prevent or cure blindness with a comparably simple cataract surgery, but many countries lack the resources to strengthen healthcare systems and implement broad cataract surgery programs reaching, in particular, the rural poor. In this paper, we analyse whether such an intervention could be cost-effective or even cost-saving for the respective health systems. We calculate the net value of the lifelong costs of cataract with and without surgery. This calculation includes direct costs (e.g., treatment, glasses, surgery) as well as indirect cost of the caregiver and the patient. We total all costs from the year of onset of cataract until death and discount the respective values to the year of onset. We define the surgery as cost-saving if the net-value of costs with surgery is lower than without surgery. If the cost per quality adjusted life year is lower than one gross national product per capita, we define the intervention as highly cost-effective. We find that the cost-effectiveness of cataract surgery depends on the age of onset of the disease and the age of surgery. If the surgery is performed with the beginning of severe impairment, even surgery of a 78-year-old patient is still cost-saving. Almost all possible constellations are highly cost-effective, only for the very old it is questionable whether the surgery should be performed. The simulations show that cataract surgery is one of the most cost-effective interventions. However, millions of people in low-income countries still have no chance to prevent or cure blindness due to limited resources. The findings of this paper clearly call for a stronger effort to reach poor and rural populations with this cost-effective service.

Keywords: Africa; blindness; cataract; cataract surgery; health economics; low-income country; visual impairment

check for updates

Citation: Flessa, S. Cataract Surgery in Low-Income Countries: A Good Deal! *Healthcare* **2022**, *10*, 2580. https://doi.org/10.3390/healthcare10122580

Academic Editor: Pedram Sendi

Received: 25 October 2022
Accepted: 16 December 2022
Published: 19 December 2022

1. Introduction

Blindness and vision impairment are a major medical and economic problem worldwide. The 'Lancet Global Health Commission on Global Eye Health: vision beyond 2020' states that some 1.1 billion people are suffering from some kind of vision impairment [1]; other organisations estimate even higher numbers [2]. Roughly half of people living with vision impairment or blindness have distance vision impairment, the other half have near vision impairment [1] with presbyopia, cataract, refractive errors and glaucoma as the main causes. The prevalence of vision impairment is much higher in low- and middle-income than in high-income countries, and, in particular, prevention and cure of eye diseases is a major issue of concern in many least developed countries [3].

Cataract is a cloudy area in the lens leading to decreased vision [4]. It is estimated that about 50% of all blindness and 33% of visual impairment are caused by cataracts [1] although a comparably simple lens replacement (cataract surgery) can highly effectively restore vision in most cases. Consequently, cataract is mainly a medical problem of low- and middle-income countries to be faced by the vulnerable population, in particular, people living in rural areas, older people, women and the illiterate [5]. With the demographic transition of these countries, it is expected that the prevalence of cataracts will increase

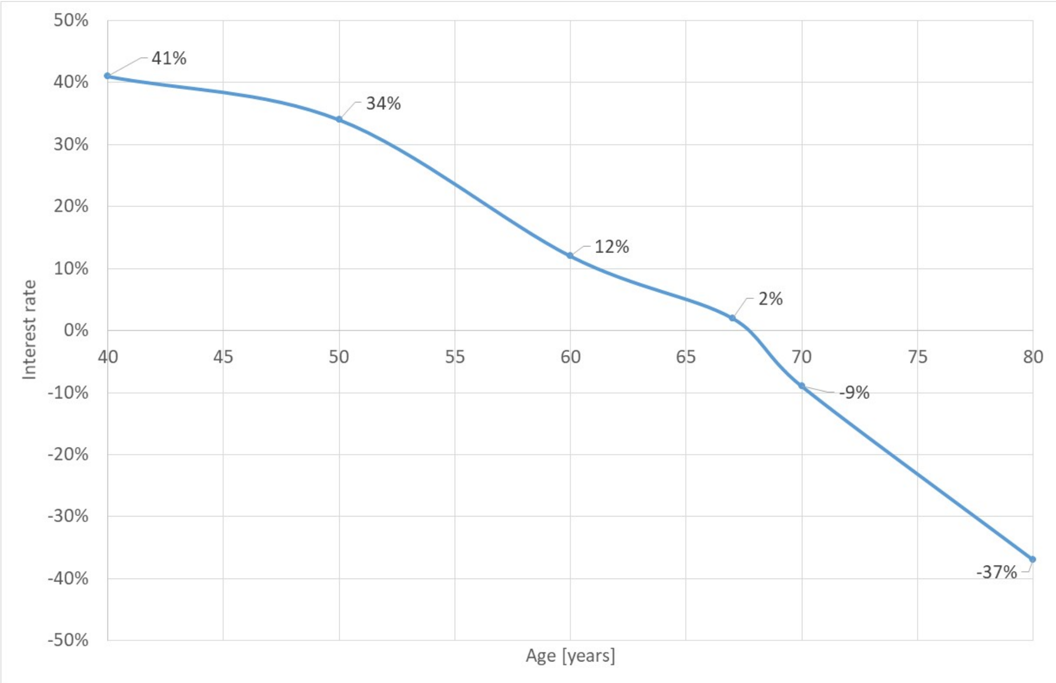

Figure 4. Return-on-investment of cataract surgery for surgery on onset. Source: own.

Finally, we have to ask for the conditions that a surgery is (1) not cost-saving or (2) not cost-effective anymore. A first answer is given by Figure 3, showing that the age of onset and the age of surgery are the most crucial conditions. The earlier a person develops cataract and is operated on, the more efficient the surgical intervention is. However, other parameters also influence cost-savings and cost-effectiveness. Table 4 shows the thresholds of parameters (for surgery in the year of onset), i.e., we change the parameters of the baseline model until the surgery is not cost-saving or (highly) cost-effective any more. The calculation is done for a person who is 60 years when mild/moderate impairment due to cataract starts and is operated on in the first year.

A first result is that the cost of treatment can be reduced to zero, but the surgery remains cost-saving. Instead of a net-value of cost-savings of 329.73 USD, the respective value is 111.76 USD. Originally, the life expectancy of a 60-year-old person is 77. If we reduce the life expectancy to 73, the surgery is not cost-saving any more. However, it remains highly cost-effective (i.e., with a cost-effectiveness ratio of less than 1 GDP p.c. per QALY) for a life expectancy of 61, i.e., even if the person will live only for one more year, it is cost-effective to perform the surgery.

The duration of the stages does also have an impact on the corridors. If the duration with mild/moderate impairment is increased to 15 years, the surgery is not cost-effective anymore, but the parameter cannot be increased so much that the intervention is not cost-effective. The duration of cataract with severe impairment will not have an impact on the corridors, i.e., even if we increase this parameter strongly, the cost-effectiveness corridor will not be left. If the GDP p.c. is reduced to 220 USD, the surgery will not be cost-saving anymore. Only for the completely unrealistic GDP p.c. of 25 USD the intervention will not be cost-effective.

A variation of the productivity loss of the patient will not have an impact on this case because the person will retire in the same year as the onset of the disease. The same is true for single variations of the productivity loss of the caregiver. Only if we reduce the productivity loss for severe impairment and blindness (e.g., 10 USD p.a. and 58 USD p.a.) the intervention is not cost-saving anymore. The pension age will not make a difference. Only if it were earlier, it would reduce the cost-savings, but the patient here is already 60 years old.

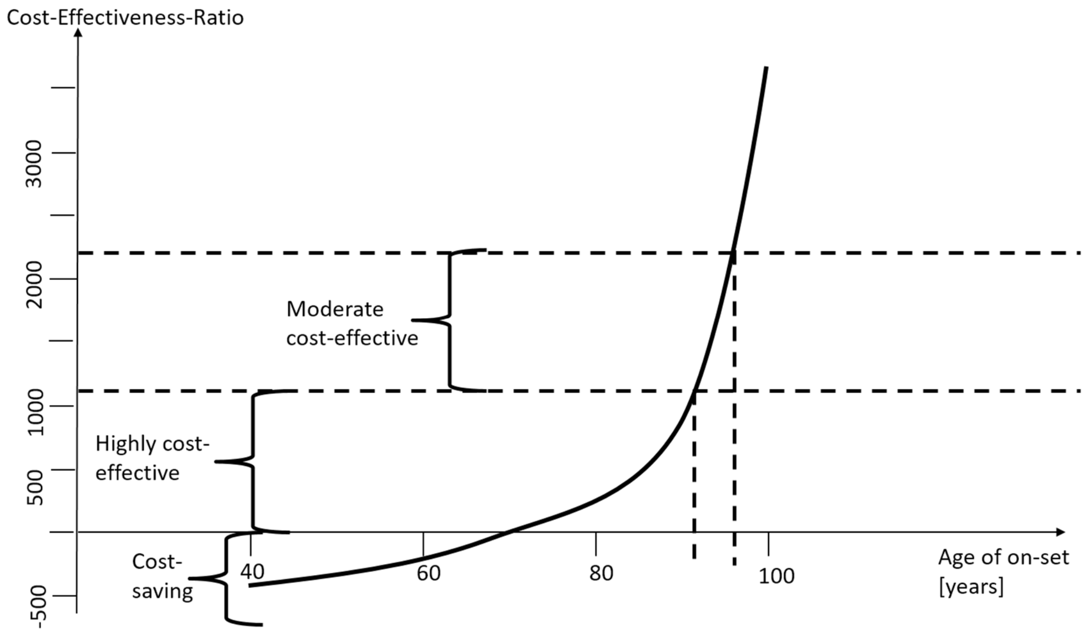

Figure 5. Cost-saving and cost-effectiveness corridor. Source: own.

Economic, social, demographic and cultural barriers lead to a situation where effective cataract surgical coverage (as the number of people in a population who have been operated on for cataract with a good outcome divided by the number of people operated on or requiring surgery) can be as low as 3.8% (Guinea Bissau) with a median of 14.8% in low-income countries [25]. Consequently, there is general agreement that there is a need to invest more into cataract surgery, and costs are not the only prohibiting factor, but an important one [26]. However, the actual cost of the respective programs and interventions are unknown or obsolete [10,27,28] for most low-income countries and respective estimates of cost-effectiveness based on these older publications are outdated themselves [29,30].

This paper gives estimates of actual direct and indirect costs and the resulting cost-effectiveness of cataract surgery in low-income countries. The respective findings are in line with other studies pointing at a high cost-effectiveness of cataract surgery. For instance, Lansingh et al. calculated the cost-effectiveness for "developing countries" as 90 to 370 international dollars per DALY averted [13] which is also demonstrating high cost-effectiveness of the intervention. However, their calculation was from the year 2007, and this clear evidence did not inspire a worldwide campaign to fight cataract in low-income countries. The evidence of our calculation and of other studies calls for a worldwide effort to end the avoidable blindness due to cataract—but this finding must be translated into a global policy.

The findings of this paper must be seen in the light of different types of limitations of this study.

- Limitations of data. There are a number of limitations of this study due to missing or unreliable data while some data might be wrong for specific situations. Although the sensitivity analyses show that the parameters might change strongly without having a major impact on the thresholds of cost-savings and cost-effectiveness, there is still a possibility that the parameters of a real situation might be very different. For instance, our model does not apply to African patients flying for cataract surgery to Arabic countries as the costs of this surgery are much higher than what we assumed here. Furthermore, costs and outcomes depend on the professionalism of services provided. Poor quality of services will result in a completely different result of the health economic analysis of cataract surgery in low-income countries. However, the vast majority of cataract surgeries in these countries are performed professionally and have a tremendous result on the visual strength of the patient and his ability to live an